

TASK – 4 WINDOWS FIREWALL SETUP AND CONFIGURATION

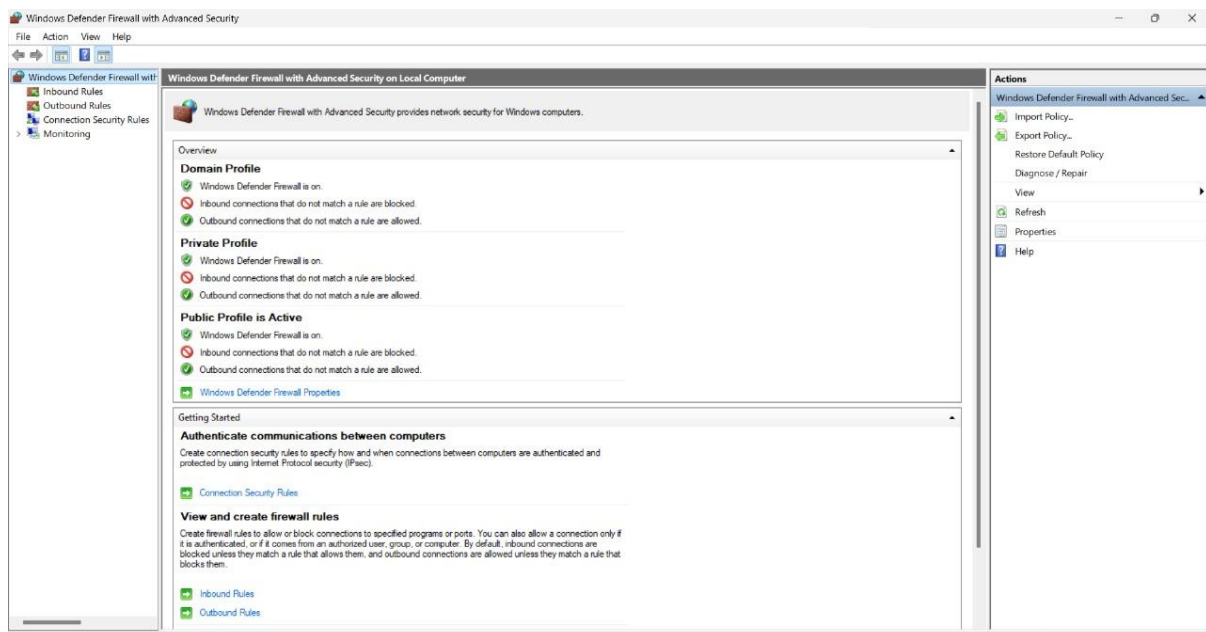
Tools used: Windows Firewall, PowerShell, Google, and ChatGPT.

INTRODUCTION:

The purpose of this task is to gain practical experience in configuring and managing firewall rules on Windows and Linux systems. Firewalls are critical components in network security, responsible for controlling inbound and outbound traffic based on predefined security policies. In this task, basic firewall operations such as allowing and blocking specific network ports are performed to understand how firewalls filter network traffic and protect systems from unauthorized access.

The task involves setting up rules to block potentially insecure ports (such as Telnet on port 23) and allow essential services (such as SSH on port 22). Testing these rules ensures that the firewall is correctly enforcing the security measures. This hands-on activity helps develop essential skills in firewall management, network traffic filtering, and security policy implementation, which are vital in cybersecurity roles.

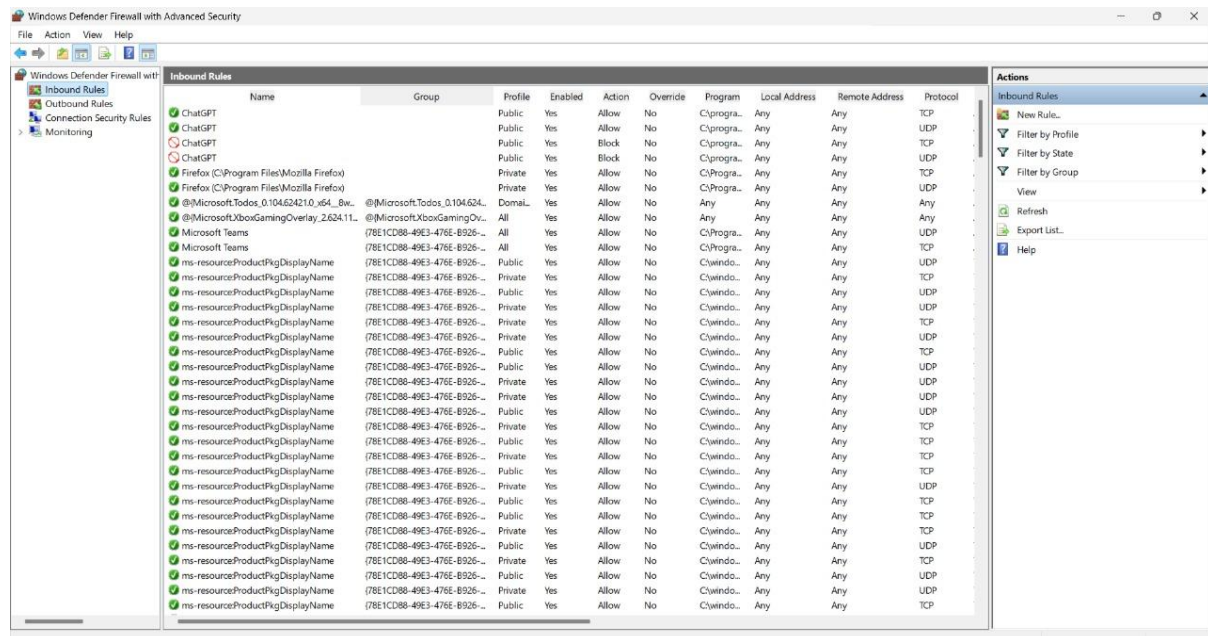
WINDOWS FIREWALL:



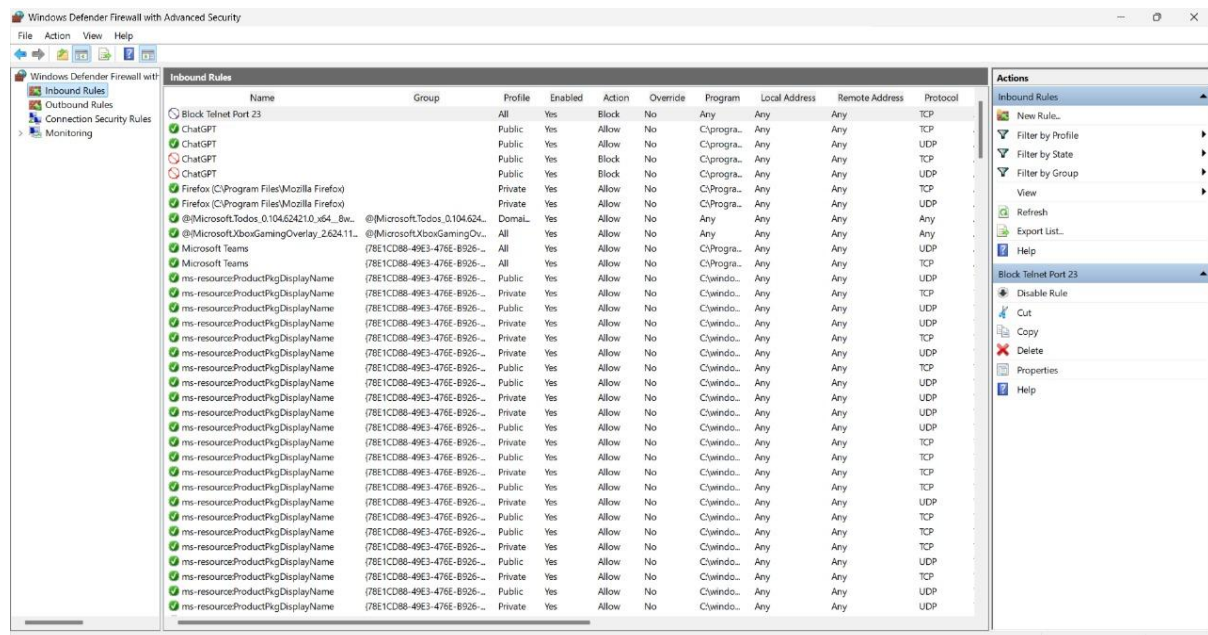
Windows Firewall Main Interface

Windows Firewall is a built-in security feature in Microsoft Windows operating systems that helps protect computers by filtering incoming and outgoing network traffic. It uses

a set of rules to allow or block specific network connections based on criteria such as ports, programs, and IP addresses. Windows Firewall helps prevent unauthorized access and protects the system from various network-based threats by controlling which applications and services can communicate over the network.



Inbound Rules View



Block Telnet Port 23 Rule

Through an easy-to-use interface or command-line tools, users and administrators can configure firewall rules to secure their devices while allowing necessary traffic.

PORTS:

Ports are numerical identifiers in computer networks that help direct traffic to specific services or applications on a device.

```
PS C:\Users\HP> netstat -an | findstr LISTENING
TCP    0.0.0.0:135          0.0.0.0:0          LISTENING
TCP    0.0.0.0:445          0.0.0.0:0          LISTENING
TCP    0.0.0.0:5040         0.0.0.0:0          LISTENING
TCP    0.0.0.0:8834         0.0.0.0:0          LISTENING
TCP    0.0.0.0:49664        0.0.0.0:0          LISTENING
TCP    0.0.0.0:49665        0.0.0.0:0          LISTENING
TCP    0.0.0.0:49668        0.0.0.0:0          LISTENING
TCP    0.0.0.0:49669        0.0.0.0:0          LISTENING
TCP    0.0.0.0:49672        0.0.0.0:0          LISTENING
TCP    0.0.0.0:49678        0.0.0.0:0          LISTENING
TCP    10.1.62.249:139      0.0.0.0:0          LISTENING
TCP    127.0.0.1:40289      0.0.0.0:0          LISTENING
TCP    [::]:135            [::]:0             LISTENING
TCP    [::]:445            [::]:0             LISTENING
TCP    [::]:8834           [::]:0             LISTENING
TCP    [::]:49664          [::]:0             LISTENING
TCP    [::]:49665          [::]:0             LISTENING
TCP    [::]:49668          [::]:0             LISTENING
TCP    [::]:49669          [::]:0             LISTENING
TCP    [::]:49672          [::]:0             LISTENING
TCP    [::]:49678          [::]:0             LISTENING
TCP    [::1]:49673         [::]:0             LISTENING
```

```
C:\windows\system32\cmd.exe  Windows PowerShell  Windows PowerShell  Windows PowerShell
Windows PowerShell
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PS C:\Users\HP> netstat -an | findstr :23
TCP    10.1.62.249:23842    74.125.130.188:443 ESTABLISHED
TCP    10.1.62.249:23845    172.188.155.25:443 ESTABLISHED
TCP    10.1.62.249:23890    49.44.194.211:443 ESTABLISHED
TCP    10.1.62.249:23891    52.168.117.170:443 ESTABLISHED
TCP    10.1.62.249:23893    52.104.7.41:443    ESTABLISHED
TCP    10.1.62.249:23895    104.18.32.47:443   ESTABLISHED
TCP    10.1.62.249:23896    104.18.32.47:443   ESTABLISHED
TCP    10.1.62.249:23898    54.197.99.182:443 ESTABLISHED
TCP    10.1.62.249:23899    54.84.9.162:443    ESTABLISHED
TCP    10.1.62.249:23902    57.144.147.32:443  ESTABLISHED
TCP    10.1.62.249:23904    49.44.62.224:443   ESTABLISHED
TCP    10.1.62.249:23905    49.44.232.225:443  ESTABLISHED
TCP    10.1.62.249:23906    49.44.62.162:443   ESTABLISHED
TCP    10.1.62.249:23907    49.44.244.226:443  ESTABLISHED
TCP    10.1.62.249:23908    49.44.63.161:443   ESTABLISHED
TCP    10.1.62.249:23909    157.240.237.60:443 ESTABLISHED
TCP    10.1.62.249:23910    49.44.241.225:443  ESTABLISHED
TCP    10.1.62.249:23911    49.44.247.164:443  ESTABLISHED
TCP    10.1.62.249:23912    49.44.243.226:443  ESTABLISHED
TCP    10.1.62.249:23913    49.44.247.98:443   ESTABLISHED
TCP    10.1.62.249:23915    107.20.252.225:443 ESTABLISHED
TCP    10.1.62.249:23916    107.20.252.225:443 ESTABLISHED
TCP    10.1.62.249:23917    98.82.75.166:443   ESTABLISHED
TCP    10.1.62.249:23918    54.192.142.105:443 ESTABLISHED
TCP    10.1.62.249:23919    54.163.127.207:443 ESTABLISHED
TCP    10.1.62.249:23920    34.235.8.204:443   ESTABLISHED
TCP    10.1.62.249:23921    52.5.196.251:443   ESTABLISHED
PS C:\Users\HP>
```

They range from 0 to 65535, with well-known ports (0-1023) assigned to common services like SSH (port 22) and Telnet (port 23). Controlling access to these ports is essential for maintaining network security and the proper functioning of services.

FIREWALL CONFIGURATION:

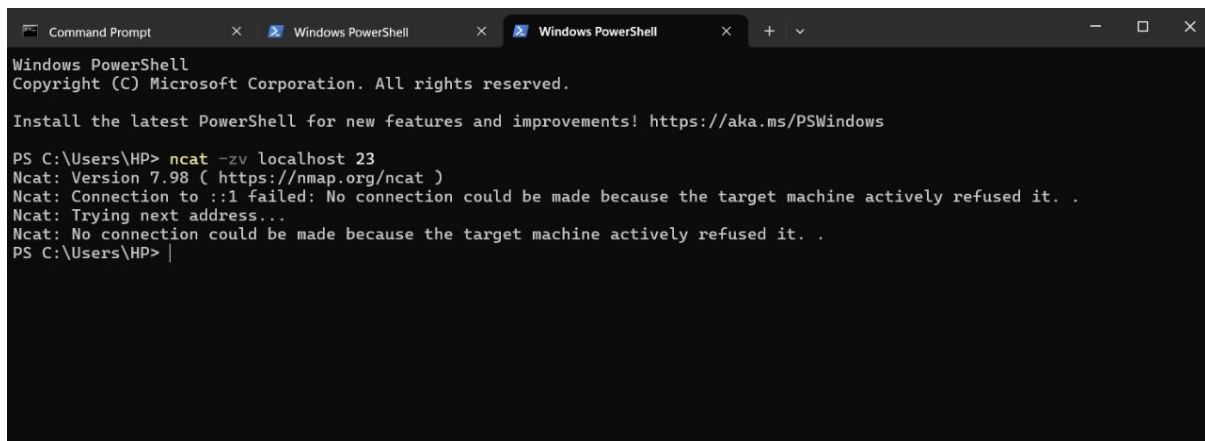
Firewall configuration involves setting up rules and policies to regulate the incoming and outgoing network traffic on a system or network. The primary goal of firewall configuration is to permit legitimate traffic and block unauthorized or potentially harmful traffic based on IP addresses, ports, protocols, and application-level data. This configuration is essential for protecting systems from cyber threats such as unauthorized access, malware, and network attacks.

Typically, firewall configuration includes creating rules to allow or block specific ports, protocols, or IP addresses. For example, blocking port 23 (Telnet) is a common security practice since Telnet transmits data in plain text and is vulnerable to attacks. Conversely, allowing ports like 22 (SSH) facilitates secure remote management.

Effective firewall configuration requires careful planning, implementation, and testing. Testing confirms that rules are working as intended, preventing vulnerabilities while maintaining necessary connectivity for users and applications. Firewalls can be managed via graphical user interfaces or command-line tools such as Windows Firewall or UFW on Linux, making firewall configuration both flexible and essential in network security management.

NETWORK TRAFFIC FILTERING:

Network traffic filtering is the process of monitoring and controlling data packets as they travel across a network. Filters are applied to allow or block specific types of traffic based on rules defined by the network administrator. These rules can be based on factors such as IP addresses, port numbers, protocols, or packet content.

A screenshot of a Windows PowerShell terminal window. The window title bar shows 'Command Prompt', 'Windows PowerShell', and 'Windows PowerShell'. The terminal text reads: 'Windows PowerShell', 'Copyright (C) Microsoft Corporation. All rights reserved.', 'Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows', 'PS C:\Users\HP> ncat -zv localhost 23', 'Ncat: Version 7.98 (https://nmap.org/ncat)', 'Ncat: Connection to ::1 failed: No connection could be made because the target machine actively refused it. .', 'Ncat: Trying next address...', 'Ncat: No connection could be made because the target machine actively refused it. .', 'PS C:\Users\HP> |'.

```
Windows PowerShell
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PS C:\Users\HP> ncat -zv localhost 23
Ncat: Version 7.98 ( https://nmap.org/ncat )
Ncat: Connection to ::1 failed: No connection could be made because the target machine actively refused it. .
Ncat: Trying next address...
Ncat: No connection could be made because the target machine actively refused it. .
PS C:\Users\HP> |
```

The main purpose of traffic filtering is to protect the network from unauthorized access, attacks, and unwanted traffic while ensuring that legitimate communications are allowed. Firewalls commonly use traffic filtering techniques to enforce security policies and maintain network safety.


```
C:\windows\system32\cmd.exe x Windows PowerShell x Windows PowerShell x + v
Windows PowerShell
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PS C:\Users\HP> Test-NetConnection -ComputerName localhost -Port 23
WARNING: TCP connect to (::1 : 23) failed
WARNING: TCP connect to (127.0.0.1 : 23) failed

ComputerName           : localhost
RemoteAddress           : ::1
RemotePort              : 23
InterfaceAlias          : Loopback Pseudo-Interface 1
SourceAddress           : ::1
PingSucceeded           : True
PingReplyDetails (RTT)  : 0 ms
TcpTestSucceeded        : False

PS C:\Users\HP> |
```

CONCLUSION:

This task successfully demonstrated the practical implementation and testing of firewall rules using Windows Defender Firewall. The objective of configuring and testing basic firewall rules to control network traffic was achieved through systematic steps of rule creation, implementation, and verification.

The task involved blocking port 23 (Telnet) to prevent insecure connections and testing the effectiveness of this rule using multiple methods including PowerShell's Test-NetConnection command, ncat utility, and telnet command. All testing methods consistently showed "Connection refused" or "TcpTestSucceeded: False" results, confirming that the firewall rule was working correctly to block unauthorized access attempts.

Through this hands-on experience, essential skills in firewall management and network security were developed, including understanding how firewalls filter traffic, creating and managing firewall rules, and validating security implementations. The task highlighted the importance of proper firewall configuration in protecting systems from potential security threats while maintaining necessary network functionality.

This practical exercise provided valuable insight into network security fundamentals and reinforced the critical role that firewalls play in maintaining cybersecurity infrastructure.
